

Cross-Attention Based Two-Branch Networks for Document Image Forgery Localization in the Metaverse

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In recent years, the Metaverse has garnered significant attention in social and Metahuman realms, showcasing substantial value and immense developmental potential through its integration of virtual and real worlds. However, this integration has also raised security concerns. For instance, in digital image forensics, the malicious dissemination of false images by wrongdoers could result in serious consequences and the propagation of misinformation. This article presented a novel two-branch network (abbreviated as CAFTB-Net) to detect and localize the forged regions of document images in the Metaverse. One branch extracts manipulation trace directly from spatial information, e.g., unnatural smears, anomalies between pixels. The other branch employs an SRM filter to transform the input image from the color domain into the noise domain, effectively extracting anomalies, such as global noise inconsistencies from the noise domain. Compared to spatial domain features, the discontinuity of forgery traces within the noise domain aids the network in authenticating the document image. The two branches extract local and global features in the forged document images. Finally, we propose a cross-attention module to fuse local spatial features and global noise features. Extensive experimental results demonstrate that the proposed network achieves F1 scores of 0.819 and 0.948 on the SACP and ICDAR datasets, respectively, with AUC scores of 0.933 and 0.764, outperforming some of the state-of-the-art algorithms.

CCS Concepts: • Computing methodologies → **Computer vision**;

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1 Introduction

With the rise of the Metaverse, we have entered a new era of digitization and virtualization, where virtual worlds are closely intertwined with the real world. The Metaverse, as a vast digital ecosystem [1, 2], offers infinite creative and interactive possibilities, attracting the participation of millions of users and businesses.

Image security is crucial in the digital age itself, but it becomes particularly prominent in the Metaverse [3–5]. In the Metaverse, people can engage in immersive interactions, such as augmented reality and brain–computer interfaces, which surpass traditional interactions based on text, images, and videos in terms of immersion and intuitiveness. Despite these advantages, the Metaverse is also vulnerable to various security and privacy risks, including personal privacy breaches, data theft, unauthorized access, and the dissemination of false information [6–8]. Within this context, users often participate in image-sharing activities to exchange personal privacy, business information, and secure confidential documents [9]. This exchange involves a wealth of crucial information. However, if malicious entities exploit specialized techniques to tamper with images within the Metaverse and disseminate falsified content, it could lead to significant risks.

As illustrated in Figure 1, users uploading photos and engaging in malicious forgery, typically through copying, pasting, or stitching alterations, could trigger a cascade of adverse effects within the internal ecosystem of the Metaverse. Therefore, it is essential to conduct research on image tamper detection to establish a more secure, authentic, and trustworthy virtual environment in the Metaverse [10].

This article focuses on document image manipulation within the Metaverse, employing deep learning techniques for secure assessment of altered images. Specifically, it introduces a dual-branch network to extract local and global features from document images. It utilizes a **cross-attention module (CAF)** to merge and filter features from both branches. Ultimately, this approach enables the localization of forged regions within manipulated document images. Such methodology significantly enhances the security of information transmission through document images within the Metaverse.

1.1 State of the Art (SOTA)

In order to detect whether an image has been forgery with or not, over the past decades, many scholars have devoted considerable effort to this research and proposed many methods for detecting and locating image forgeries [11–14]. Zhang et al. [15] applied **photo-response non-uniformity (PRNU)** and a deep fully connected neural network to locate tampered regions in images. This method first calculated tampering probability maps at different scales based on PRNU correlation, then utilized the neural network’s powerful data adaptability to integrate multi-scale information, reducing false-positive rates. PRNU is caused by sensors and is considered as device fingerprinting. Different device fingerprints have different characteristics, enabling the identification of inconsistent regions in the target image using this feature. Wen et al. [16] proposed a method to estimate

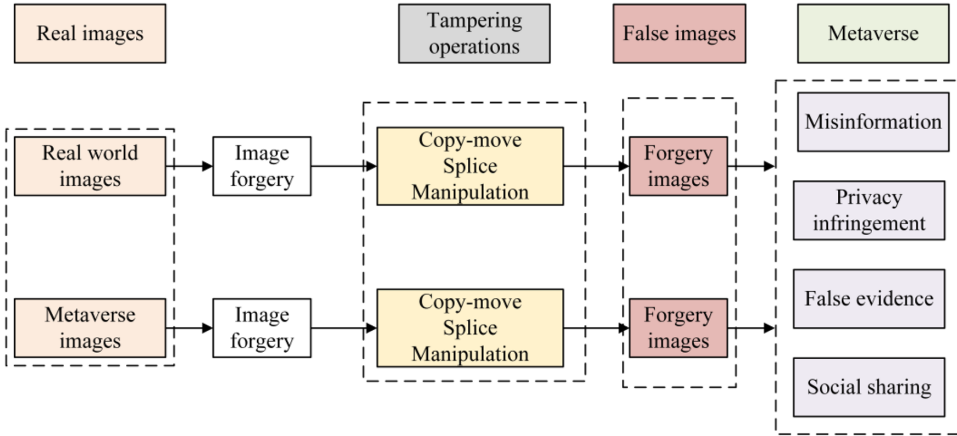


Fig. 1. The hazards of forgery images in the Metaverse.

contrast enhancement operations from a single image. This method utilized the fluctuation changes in statistical properties brought about by contrast enhancement, restored the original pixel histogram through an iterative algorithm and the applied contrast enhancement, effectively handling noise and disturbances caused by compression. Using this method, regions of spliced images transformed by different contrast enhancements can be detected. Given the achievements of deep learning in numerous computer vision tasks like semantic segmentation and object detection, numerous techniques based on deep learning have been devised for the detection and localization of image forgeries. Most of these methods are based on **Convolutional Neural Network (CNN)** or transformer networks. For instance, Niloy et al. [17] proposed CFL-Net based on CNN, which uses a two-branch network and constraint learning to solve the problem of easy overfitting of traditional forgery localization methods. To address the issue of low accuracy, Sun et al. [18] proposed a transformer network for forgery region localization called **Edge-Enhanced Transformer (ET)**, it uses a multi-layer transformer layer and Sobel operator to learn the features and a self-attention-based augmentation module to highlight the edge artifacts, which enables a significant improvement in the localization performance. Wang et al. [19] may not use the CNN structure employed by most methods to capture image-level or pixel-level forged images but instead modeled object-level visual inconsistency information with the advantage of a visual transformer, thus providing a novel idea for image forgery detection. The method achieves SOTA results on commonly used image forgery datasets. Lin et al. [20] proposed a two-branch network, EMT-Net, by combining CNN and transformer networks' properties. This network can learn traces left by various forgery operations, and the network's learning ability and robustness are dramatically improved through enhancement and supervision strategies for edge traces.

However, most of the forgery detection and localization methods mentioned above focus on natural images rather than document images. Based on the differences between natural and documented images, approaches that can detect natural images are not necessarily applicable to the detection of document images. Figure 2 displays the difference between a faked natural image and a real-world document image. Natural scene image forgery usually refers to operations such as copy and paste, splicing, and removal, and the forgery objects are focused on objects with more obvious forgery traces. Document image counterfeiting operations focus on text, patterns, and after careful post-processing, almost invisible counterfeit traces, in addition to natural image counterfeiting operations, including content generation, filtering, and so forth. In addition, compared with natural



Fig. 2. Forgery of natural image and document image.

images, document images usually contain sensitive and critical information. The use of forged document images can have serious consequences, such as false evidence, illegal theft, fraud, and reduced social trust, given the unique characteristics of document-type files, such as having a complex background texture, no noticeable closed edges. It makes the traditional forgery detection methods less effective in applying document-type files. Clearly, this falls short in addressing the complexities and variability within the Metaverse. Hence, the immediate priority is to conduct in-depth research and enhancements specifically targeting the forgery of document images. This aims to elevate the effectiveness and practicality of detection algorithms within the Metaverse.

Forensic research on document images has attracted the interest of academics and corporations alike. Some detection methods have been proposed one after another, and Alibaba has organized document text class challenges several times [21, 22]. Cruz et al. [23] used Local Binary Pattern to compute texture features on the forged region and then used Support Vector Machine to classify the image blocks as either original or forgery. James et al. [24] used Optical Character Recognition to extract features such as height, width, and spacing of a single character and used random forest to classify characters. Garima Jaiswal et al. [25] proposed to tackle the problem of detecting ink mismatches in document forgeries, specifically in hyperspectral document images. It introduces an unsupervised deep learning method that utilizes **Convolutional Autoencoders (CAE)** for deep feature extraction and **Logistic Regression (LR)** for classification, named CAE-LR. This novel approach was evaluated using the UWA Writing Ink Hyperspectral Images dataset, demonstrating its superiority over several machine learning algorithms and state-of-the-art methods in accurately detecting ink mismatches. The findings underscore the potential of CAE-LR in enhancing forensic document analysis by efficiently handling spectral features for forgery detection. However, these methods mostly rely on manually extracted features and are only effective for certain types of tampering operations, often failing to achieve satisfactory performance in complex scenarios.

Most of these methods rely on manually created features and are often unable to satisfactorily address complex forensic scenarios. Xu et al. [26] proposed a two-branch network for document image forgery localization, which extracts spatial domain features such as edge differences and anomalous features between neighbouring pixels. This method achieved better performance in region-level localization. Sun et al. [27] introduced an effective MFAN network for detecting forged certificate images. By integrating feature maps from low-level convolutional layers, the network

preserves more information and enhances the localization of smaller forgery objects, mitigating potential failures. Liang et al. [28] argued that forgery images may undergo additional processing or transmission through distorted channels, potentially degrading the network's detection performance. Therefore, they introduced a robust document image forgery localization approach specifically designed to combat image blending, significantly enhancing the accuracy of forgery localization. Zheng et al. [29] proposed a coding–decoding network to distinguish between tampered and desensitized areas in original images, as both tampering and desensitization operations introduce additional traces. Specifically, they employed ConvNext as the backbone network and integrated multiple **squeeze-and-excitation (SE)** attentions to suppress image content and enhance the network's focus on the target. Qu et al. [30] proposed a document tampering detector named **Document Tampering Detector (DTD)**, which is used to capture finer clues for tampered text detection in complex scenarios. Specifically, by extracting features from both the frequency domain and spatial domain, and by designing multi-level iterative decoders that mimic the scaling operations people use when analyzing images, they fully utilize features of different sizes to predict more accurate results. However, these approaches still focus on block-level forgery detection, resulting in less precise localization of tampering in document images.

In general, the prevailing focus has been on detecting forgery within natural images, relegating studies pertaining to document images, especially within the context of document image security in the Metaverse, to a relative minority. The metaverse is a virtual digital world, where its unique environment and application scenarios may lead to different document image tampering methods compared to traditional environments. The metaverse environment is highly virtualized and interactive, posing new technical challenges for document image tampering detection. For instance, in virtual worlds, users can easily edit, copy, and distribute document images, which may increase the complexity of detection. Therefore, considering the metaverse as a specific domain facilitates deeper research and exploration into the technical challenges presented within it. Currently, research on forgery detection in document images still faces challenges such as coarse localization results and insufficient localization accuracy. Detecting and pinpointing forgery in document-like images represents an inherently arduous task, particularly within the complex environment of the Metaverse [31]. Forgery detection localization of document-like images is a very challenging task with two main challenges in the Metaverse: (1) Due to the complexity of document image features, forged regions typically consist of only a few pixels, resulting in a severe imbalance between positive and negative samples. Additionally, forged regions bear a high degree of similarity to authentic regions. Current methods for document image forgery detection struggle to effectively discern between forged and authentic pixels through feature extraction solely from the spatial domain. (2) Extracting features from different domains, such as color and noise domains, and integrating them can effectively enhance the network's discriminative ability. However, seamlessly fusing features from diverse domains without losing tampering information poses certain challenges. The above challenges greatly limit the performance of document image forgery detection methods.

1.2 Contribution

To enhance the security of document images in the Metaverse and foster a safe, reliable virtual environment, this article introduces a novel approach: the **Cross-Attention Based Two-Branch Network (CAFTB-Net)** for detecting document image forgery within the Metaverse context. CAFTB-Net is adept at pinpointing the specific locations of alterations at the pixel level in document images. The proposed model utilizes features from both the spatial domain and the noise domain. The former feature, based on the original image input, is extracted using the **spatial information extraction branch (SIEB)**, which comprises components with convolutional structures to extract spatial information. As for the latter, we utilize an **Spatial Rich Model (SRM)** filter to transform

the original image into a noise image and employ the **noise feature extraction branch (NFEB)** to capture global noise features from the noise domain. Specifically, SIEB aims to extract local features from the edge neighborhood, aiding in detecting inconsistencies between forged regions and their surroundings. The global features extracted by NFEB enhance the localization capability of the entire image. Finally, we designed a fusion module named CAFM with a cross-structure and attention mechanism to effectively integrate spatial and noise features. We compared CAFTB-Net with other forgery detection methods for both natural and document images. The proposed method achieved an 11% and 2% increase in **area under the ROC curve (AUC)** and F1 scores, respectively, over the second-best method on the SACP dataset. On the ICDAR dataset, it outperformed the second best by 14% and 2% in terms of AUC and F1 scores, respectively. These results demonstrate the significant advantage of our proposed method in detecting forgeries based on document images. The primary contributions of this study are outlined below:

- (1) A two-branch document image tampering localization network is proposed for the Metaverse. One branch is a SIEB to capture forgery traces directly from document images. The other branch called the NFEB uses an SRM filter to convert the input image from the color domain to the noise domain to mine anomalies such as global noise inconsistencies from the noise domain.
- (2) A CAFM is proposed by using channel attention and residual spatial attention to recover and enhance the features extracted by the SIEB and NFEB. CAFM can capture more accurate and critical information in the feature fusion process, thus improving the performance and robustness of the model.
- (3) We conducted comprehensive experiments on two competition datasets, demonstrating that our approach surpasses several existing methods for document image forgery localization.

1.3 Organization

The remaining sections of this article are structured as follows: Section 2 delves into the preliminary. Section 3 offers a comprehensive overview of the proposed methodology. Extensive experimental results, demonstrating the superiority of the proposed approach, are presented in Section 4. Finally, Section 5 draws the conclusions of this study.

2 Preliminary

2.1 Document Image Forgery Detection and Localization

Document image forgery detection refers to determining whether a document class image has been forged, and document forgery localization requires locating the forged region in the document image. The forgery of document images is primarily concentrated on the seal, signature, and other critical information, and some complex post-processing operations are used, such as changing the shape, rotation, JPEG compression and others, which poses significant challenges to document image forgery localization. Most of the current deep learning-based document image forgery detection methods are based on the text block or image block approach [24, 26, 32], such as Nandanwar L [32], James [24], Xu [26], and others. These methods first divided the image or the text region in the image, then calculated the similar features of each chunk separately, and finally classified them using a CNN network. Although these methods have achieved good localization results, there is an obvious block effect in the localization of forged areas, resulting in insufficient localization accuracy. Overall, some of the existing approaches rely on manually extracting features, which can only address specific types of tampering and may struggle to perform well in complex

scenarios. Other methods focus solely on detecting tampering at the image block level, resulting in inadequate localization accuracy for document images. Additionally, the network's performance is influenced by the size and number of image chunks, with overly large or numerous chunks requiring increased computational resources. These approaches primarily capture semantic feature differences in the spatial domain. However, forgery operations typically leave traces in the noise domain, posing challenges for spatial domain feature extraction. In this study, noise domain features are introduced and integrated with spatial domain features to enhance the network's capability in identifying tampering traces and achieving improved accuracy in document image forgery detection.

The proposed CAFTB-Net utilizes an end-to-end training approach without any pre-processing or post-processing operations. This permits finer pixel-level localization performance. With this network, the forged regions in document images can be accurately localized, which solves the problems of the chunking effect and insufficient localization accuracy in some existing methods. In addition, due to the end-to-end design of the network, there is no need to manually select the chunk size and number, which reduces the demand for computational resources and makes document image forgery localization more efficient and reliable.

2.2 Two-Branch Networks in Forgery Detection

Most existing localization methods for image forgery detection use CNN, which are feed-forward neural networks with deep structure and convolutional computation. CNN are excellent at extracting local features but require learning global representations. Within neural networks, global feature extraction can be achieved by expanding the receptive field, as opposed to focusing solely on local feature extraction [33]. However, a sizeable receptive field also implies an enormous computational effort. In addition, global attention mechanisms [34, 35] can capture long-range dependencies. However, local feature representation may deteriorate if convolution is not properly integrated with the attention mechanism.

The transformer architecture derives a comprehensive image representation through an encoder-decoder network featuring global self-attention, uniquely reliant on the attention mechanism while eschewing recursion and convolution, as indicated in [36]. Originally conceived for natural language processing, the transformer model has been successfully adapted to vision-related tasks, with transformer blocks introduced in [37] showcasing commendable performance in forgery detection tasks. As mentioned above, the ET network [18], which first splits the image into blocks and then uses the transformer layer to extract the image block features, ET-Net got good performance on spliced type of forged images. However, the transformer network focuses more on the global features of the image. It needs to capture the local details efficiently, which may lead to insufficient accuracy in localizing the forgery regions.

The dual-branch network structure has been increasingly used for forgery detection tasks, where scholars are interested in extracting forgery features from the other branch that can complement each other. Xu et al. [26] designed a dual-stream network for forgery localization of document images, where the spatial information extraction network is used to extract spatial information, and the other branch, the correlation feature extraction network, suppresses the image content using multiple filters that capture the correlation between pixels. Wang et al. [38] proposed a forgery text image detection method based on RGB and frequency domain relationship modeling, which enhanced the network's ability to discriminate forgery textures by introducing frequency domain information and using a global RGB-frequency domain relationship module to model the relationships between different text images to improve the detection accuracy.

In this study, to extract more discriminative features for document image forgery localization, we employ both CNN and transformer architectures to extract local and global features, respectively.

The local features are extracted by CNN, while the global representation is obtained through the transformer. These features are effectively fused in the CAFM, endowing CAFTB-Net with the structural advantages of both CNN and transformer, thereby maximizing the preservation of both local and global feature representation capabilities.

2.3 Attention in a Forgery Detection Task

In the realm of computer vision, deep learning models frequently leverage attention mechanisms to bolster feature representation. Researchers have consistently elevated the performance of diverse baseline models by introducing attention modules. For instance, Woo et al. [39] innovatively integrated channel and spatial attention sequentially in their Convolutional Block Attention Module module, resulting in heightened channel attention efficacy. Additionally, Hu et al. [40] introduced the SE attention mechanism, which addresses interdependencies among channels by infusing global information into each channel through contextual cues. Self-attention, renowned for its capacity to capture global information, finds widespread utility in the field of natural language processing. In the forgery detection task, Liu et al. [41] proposed a two-branch network to extract RGB and noise features separately. They constructed a positional attention block to study the correlation of the two features and fused them into a unified feature map. GCA-Net [42] utilizes an advanced attention mechanism that boosts image forgery detection and localization. It begins with Global Context Pooling, capturing global relationships between pixels to better discern altered regions. Then, Attention Gating selectively focuses the network's attention on the most pertinent features, effectively filtering out irrelevant background noise. This process not only sharpens the detection of manipulated areas but also improves the network's overall precision and efficiency in identifying forgeries.

In this article, we introduce the CAFM, beginning with the fusion of features extracted by the CNN branch and the transformer branch, which operate in distinct domains. CAFM proficiently combines the local features derived from the CNN with the global features obtained from the transformer, thereby significantly enhancing the network's localization performance. Experimental results from this module are shown in the ablation experiments in the experiments section.

3 Proposed Method

3.1 Overall Structure of the Model

The Metaverse serves as a digital mapping of the objective world, representing a digitized and networked manifestation of objective reality. Hence, in the Metaverse realm, the ability to objectively reflect authentic information from the real world and effectively identify malicious forgeries is of paramount importance. The primary goal of this article is to achieve pixel-level localization of forged regions within document images in the Metaverse. Since document images usually have more regular textures than natural images, objects in the images do not have noticeable closed edges and other features, using traditional CNN modeling methods may not always be applicable. Therefore, in this article, we discard the design idea of using the CNN but combine CNN and transformer to propose a new dual-stream network structure to solve the problem of document image forgery detection and localization. The network consists of a pre-processing block, a CNN constitutes a SIEB that extracts local features, and a transformer network constitutes a NFEB that extracts global features. The SIEB captures the forged traces of the original image from the null domain, including unnatural edges, brightness inconsistencies, and so forth. The other NFEB converts the image to the noise domain through an SRM filter and then mines anomalies such as global noise inconsistencies from the noise domain. Next, the features of the two streams are fused by the CAFM and then up-sampled to the size of the original image by the decoder. Finally, each pixel is judged by the classification layer to determine whether it is a forgery. The specific flow of the algorithm is shown in Figure 3.

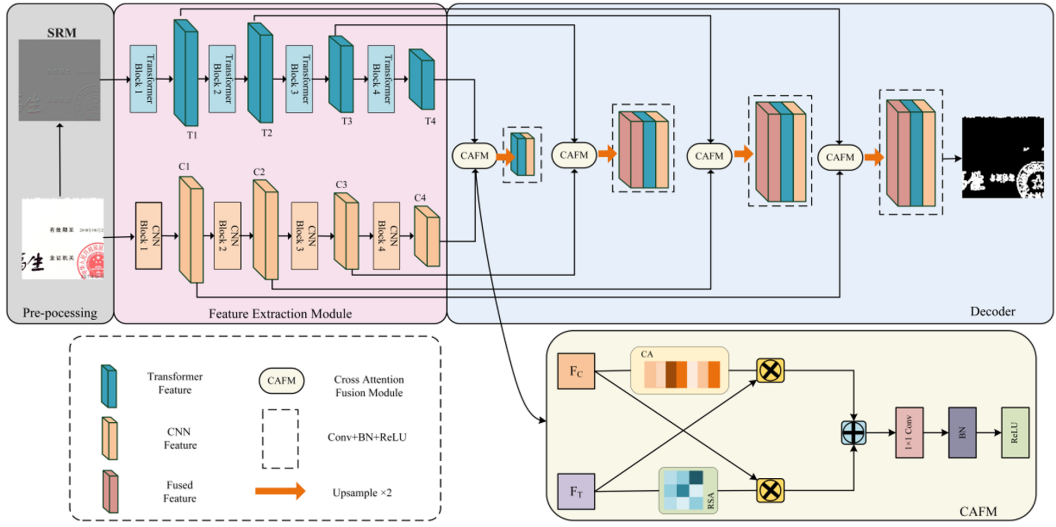


Fig. 3. Overall architecture of the proposed CAFTB-Net.

$$\frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 2 & -4 & 2 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \frac{1}{12} \begin{bmatrix} -1 & 2 & -2 & 2 & -1 \\ 2 & -6 & 8 & -6 & 2 \\ -2 & 8 & -12 & 8 & -2 \\ 2 & -6 & 8 & -6 & 2 \\ -1 & 2 & -2 & 2 & -1 \end{bmatrix} \quad \frac{1}{2} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Fig. 4. The three noise filters in the pre-processing stage are used for noise feature extraction in the noise domain.

In the following subsections, the SIEB, the NFEB, and the proposed CAFM are presented in detail.

3.2 Pre-Processing Block

The pre-processing module converts the image from the color domain to the noise domain using an SRM filter. SRM is a high-pass filter that enhances the high-frequency content in the input image and can be used as an input to the noise branch and help to detect forged regions [43]. In this study, noise is represented by calculating the residual between the actual value of the pixel and the interpolated estimate of the neighboring pixel values. As shown in Figure 4, we used three different kernels, each with weights. These kernels consist of a 3×3 rectangular filter, a 5×5 rectangular filter, and a second-order filter kernel that produces three channels of output.

The noise feature map produced after the SRM layer emphasizes local noise rather than image content and reveals forgery artifacts not visible on the RGB channels. The noise feature map can be used directly as input to the transformer branch.

3.3 SIEB

Images within the Metaverse, serving as digital representations, exhibit diverse functionalities and forms, enriching user interactions and experiences within virtual environments. Pertinently, alterations in image content directly influence spatial information, thus allowing the detection of forged documents through the identification of anomalous spatial traces. Several preceding methods for image forgery localization [17–20] have also aimed to capture forgery traces through

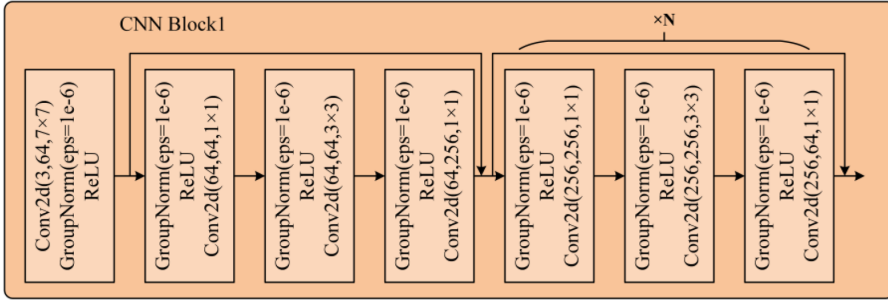


Fig. 5. SIEB. Conv Block 1 is shown in the figure.

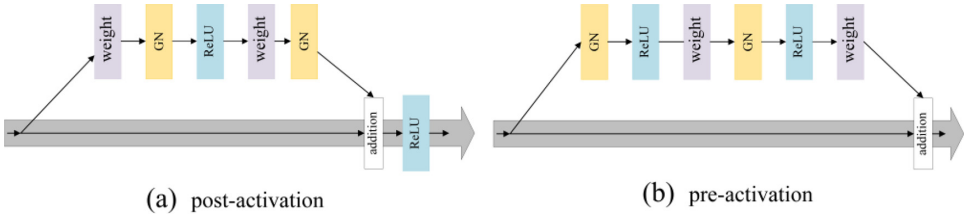


Fig. 6. Structural diagrams of the two activation approaches. (a) Post-activation method and (b) structural approach used in this article.

spatial information. The SIEB proposed in this article comprises four convolutional blocks. The features extracted by SIEB exhibit a feature pyramid structure, characterized by a reduction in feature map size as the network's depth increases. See Figure 5 for detailed information on SIEB. The first Conv block of SIEB contains a pre-processing layer and four bottleneck layers. The pre-processing layer incorporates a 7×7 convolutional layer along with **group normalization (GN)** and ReLU activation functions. Each bottleneck layer is composed of a sequence that includes a 1×1 downward-projected convolutional layer, a 3×3 spatial convolutional layer (serving as the primary feature extractor), and a 1×1 upward-projected convolutional layer. Following each convolutional layer, we insert a **batch normalization (BN)** layer and apply a ReLU activation function to augment the network's non-linear capabilities. Additionally, a residual connection is established between the input and output of each bottleneck layer. The output feature size of Conv Block 1 is $64 \times 128 \times 128$. In subsequent Conv Blocks, the pre-processing layer is removed. As the network deepens, the number of bottleneck layers in each Conv Block is gradually increased to extract progressively higher-level and semantically rich feature representations. In the setup of this article, the number N of bottleneck layers in each Conv Block is [1, 3, 4, 9] respectively. By using different numbers of bottleneck layers, SIEB can take advantage of both shallow and deep features and use them for subsequent forgery detection and localization tasks. The second Conv block contains three bottleneck layers with an output feature size of $256 \times 64 \times 64$. The third and fourth Conv blocks contain four and nine bottleneck layers, respectively, with output feature sizes of $512 \times 32 \times 32$ and $1024 \times 16 \times 16$, respectively.

Note that before each convolution, a GN layer is used to normalize the weights and use the ReLU activation function in a way called pre-activation [44]. Figure 6 shows that pre-activation is easier to train and has better generalization capabilities than the original post-activation. The activation of pre-activation is used in all the bottleneck layers in this article.

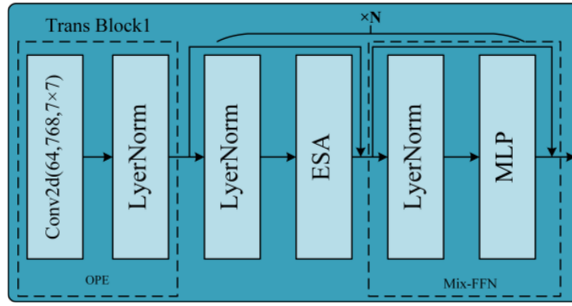


Fig. 7. Transformer Block 1 is shown in the figure. Trans block 1 refers to a block of code in the transformer network, LayerNorm represents layer normalization, ESA denotes Efficient Self-Attention, MLP refers to Multilayer Perceptron.

SIEB can extract rich feature representations from the input image. These features can capture local features in the image, including color, texture, shape, and so forth, and thus help to distinguish the difference between the forgery region and the original region. In addition, by using different numbers of bottleneck layers in different Conv blocks, the semantic information extracted by the network is gradually deepened, and this hierarchical feature extraction strategy helps to improve the network's perceptual ability and robustness, enabling it to better cope with document image forgery situations.

3.4 NFEB

In the Metaverse, as images exist in digital form, they are also subject to the influence of noise. To effectively harness the forgery traces left behind by the manipulation process, we employ NFEB to capture anomalous features related to noise inconsistency in the noise domain. This branch mainly uses the transformer network; CNN performs effectively in extracting local features, while the transformer is more skillful in extracting global features. Therefore, we use a transformer to extract features in the noise domain. The transformer branch is mainly designed regarding the structure of the SegFormer [45] network, which consists of four transformer blocks, as shown in Figure 7. The noise map after SRM filter processing is first divided into multiple patches of size 4×4 . Since CAFTB-Net is a pixel-level prediction network, it must predict every pixel in the image. In contrast to the size 16×16 patches used in ViT, smaller patches facilitate the forgery detection task. Each transformer block has the same structure and contains three modules: **Overlap Patch Embedding (OPE)**, **Efficient Self-Attention (ESA)**, and **Mix Feed-Forward Network (Mix-FFN)**, and we use lyernorm to normalize features between each module to improve training speed and stability.

OPE refers to the segmentation of input noise images into overlapping small patches, which are then fed into the transformer model as input. This approach allows for overlapping regions between adjacent patches, enabling the model to capture more information and context in these overlapping regions. OPE is designed to produce CNN-like multi-scale feature maps while avoiding the loss of continuity of patch boundaries. Unlike traditional patch cutting, OPE uses overlapping cuts to maintain the continuity of patch boundaries when cutting patches. A multi-scale feature map of size $\frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times C_i$ is obtained from the input image of $H \times W \times 3$, where $i \in 1, 2, 3, 4$. The computational workload of the network is primarily concentrated in the self-attention mechanism layer. The ESA employed in the trans block of this article reduces computational complexity by employing sequence reduction, compared to standard Self-Attention, effectively lowering the consumption of

computational resources. The reduction rate is marked as R and the computational formula for ESA is as follows:

$$K' = \text{Reshape} \left(\frac{N}{R}, C \cdot R \right) (K) \quad (1)$$

$$K = \text{Linear} (C \cdot R, C) (K'), \quad (2)$$

where K denotes the reduced sequence, and the $N \times C$ feature map is first transformed by the reshape operation into $\frac{N}{R} \times (C \cdot R)$ size of feature map. $\text{Reshape} (\frac{N}{R}, C \cdot R)(K)$ refers to reshape K into $\frac{N}{R} \times (C \cdot R)$ shape. The fully connected layer is then used to reshape $\frac{N}{R} \times (C \cdot R)$ feature maps into $\frac{N}{R} \times C$. $C \cdot R$ denotes the dimension of the input and C denotes the dimension of the output. Therefore, the dimension of the new K is $\frac{N}{R} \times C$. It follows that the complexity of self-attention will be reduced from $O(N^2)$ to $O(\frac{N^2}{R})$. This article has a total of four transformer blocks, and the parameter R is set to $[64, 16, 4, 1]$ from transformer block 1 to transformer block 4. The document image forgery localization task is pixel-level classification, and the positional coding has a negligible effect on pixel-level classification. Therefore, using a convolution through 3×3 , it is possible to express the positional relationships between patches dynamically. We use a 3×3 convolution to blend the MLP into a **feed-forward network (FFN)** to form Mix-FFN. The computational procedure of Mix-FFN is as follows:

$$X_{\text{out}} = \text{MLP}(\text{GELU}(\text{Conv}_{3 \times 3}(\text{MLP}(X_{\text{in}})))) + X_{\text{in}}, \quad (3)$$

where X_{in} refers to the output of features from the Self-Attention module, X_{out} refers to the output of the transformer block, $\text{Conv } 3 \times 3$ denotes the convolution with a convolution kernel size of 3, GELU is the activation function, and MLP is the multilayer perceptron. The main purpose of Mix-FFN is to introduce more flexibility and diversity into the transformer, enhancing the model's ability to perceive features at different scales and thereby improving its performance.

Similar to the SIEB, to obtain different features in the shallow and deep layers, different blocks of the transformer branch are varied in different depths, mainly manifested in the number of repetitions of ESA and Mix-FFN. The NFEB contains a total of four transformer blocks, and the number of repetitions N of ESA and Mix-FFN in each block is set to $[3, 6, 40]$, respectively. The features extracted from the transformer branch are one-dimensional vectors. To further integrate with the features from the SIEB in the subsequent decoder, we perform a feature alignment operation on the output of the features from each transformer block. The final size of the output feature maps from transformer block 1 to transformer block 4 are $64 \times 64 \times 64$, $256 \times 32 \times 32$, $512 \times 16 \times 16$, and $1,024 \times 8 \times 8$, respectively.

3.5 CAFM

In order to effectively detect traces of forgery, it is essential to capture intricate counterfeit details within the features. Nonetheless, downsampling operations within the encoder can substantially degrade the quality of forgery features extracted from these layers. Since forgery regions in document images are typically located in the foreground, treating all pixels uniformly and without differentiation can lead to false detections. In addition, the feature maps extracted by these two branches are significantly different because they come from the color and noise domains. Therefore, how to effectively fuse the features extracted by the two-branch network is also an integral part of the research. To address this problem, a CAFM is designed in this article, as shown in Figure 8.

CAFM consists of two attention modules: channel attention and residual spatial attention using residual connections. Channel attention is used to adjust the weights of individual channels adaptively. In contrast, spatial attention combines the operations of global maximum pooling and global average pooling to capture global contextual information. We use residual concatenation to fuse the input and spatial attention features to preserve the detailed information. The structure of

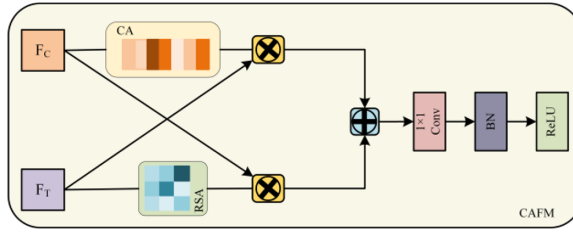


Fig. 8. Structure of the CAFM.

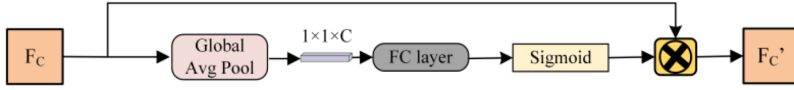


Fig. 9. Structure of channel attention.

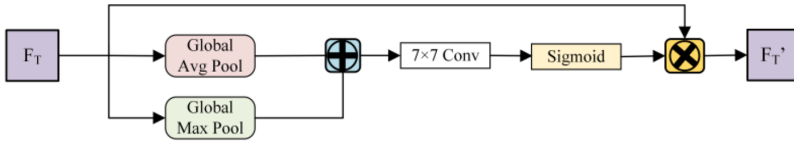


Fig. 10. Structure of RSA.

channel attention and **residual spatial attention (RSA)** is shown in Figures 9 and 10. Finally, a cross-aggregation operation is used to fuse the features of the two branches to recover and enhance the features extracted from the SIEB and NFEB branches.

First, the channel attention module [46] computes channel-wide attention maps by considering correlations between different channels. In this article, we use the features of SIEB to compute the channel attention features.

$$CA = \text{Sigmoid}(\text{FC}(\text{GAP}(F_C))), \quad (4)$$

F_C denotes the features of the SIEB, and GAP denotes global average pooling, two 1×1 convolutions are included in FC, the GAP output features are downsampled, and finally, the Sigmoid activation function is used, and the weight range is distributed between 0 and 1. Next, RSA is used for the features of the NFEB, which helps to highlight the critical features at each spatial location in the noise domain by weighting the attention to different spatial locations of the feature map. The computational procedure is as follows:

$$SA = \text{Sigmoid}(\text{Conv}_{7 \times 7}(\text{GAP}(F_T) + \text{GMP}(F_T))) \quad (5)$$

$$RSA = F_T \times SA, \quad (6)$$

F_T represents the features extracted from the NFEB. We use GAP for global average pooling, GMP for global spatial pooling, and $\text{Conv } 7 \times 7$ to indicate a convolutional kernel size of 7×7 . Subsequently, the feature maps from the two branches undergo fusion through cross-multiplication. This process reinforces the global representation within the CNN branch and amplifies the local details within the transformer branch. Following fusion, the features are further refined through a 1×1 convolution, a BN layer, and the application of a ReLU activation function.

By combining channel attention and residual space attention, CAFM can fully utilize the features extracted from SIEB and NFEB, while enhancing the representation and discrimination of these features.

4 Experimental Analyses

4.1 Experimental Setup

Datasets. The current widely used image forensics datasets are NIST-2016 [47], Columbia [48], and CASIA [49]. However, the image contents of the above datasets are mostly natural landscapes, buildings, or people and need more focus on document images. In 2020, Alibaba and other organizers held the first adversarial attack competition [18] against document image forgery called **Security AI Challenger Program (SACP)** Phase V: Adversarial Attack Competition on Forged Images. In this competition, an official document image forgery dataset that approximates real scenarios is provided, named the SACP dataset in this article; 2023 Tianchi and other organizers held (ICDAR 2023 DTT in Images 2: Text Manipulation Detection) Text Image Tampering Detection [22]. The dataset provided in this competition is more complex. It contains images taken from electronic devices such as poster images, invoices, books, and many digital images such as screenshots of chat logs. Since digital images such as chat logs are forged in a way that is too different from document images, in this experiment, the images of chat log screenshots have been removed, and only images such as documents, invoices, and books have been used for training and detecting the model of this article.

The original datasets of SACP and ICDAR 2023 contain 2,005 and 3,800 images, respectively. We augmented the two datasets to 11,320 and 12,300 images using data augmentation techniques such as rotation, mirror flipping, and cropping. Each dataset is divided into training set validation set and test set in the ratio of 8:1:1.

Evaluation Metrics. Image forgery localization is a pixel-level binary classification problem. This subsection uses three evaluation metrics including AUC, F1, and IOU. The formulae for F1 and IOU are as follows:

$$F1 = \frac{2TP}{2TP + FN + FP} \quad (7)$$

$$IOU = \frac{TP}{TP + FN + FP}. \quad (8)$$

In the context of forgery detection tasks, TP denotes the count of pixels accurately identified as belonging to the tampered region, i.e., the number of pixels correctly recognized as forgeries. FN represents the count of pixels within the forgery region that were erroneously classified as non-tampered, meaning the pixels that were not correctly identified as forgeries. FP signifies the count of pixels mistakenly classified as belonging to the tampered region, implying the incorrect classification of unaltered regions as forgeries. These metrics are commonly employed to assess the performance of forgery detection algorithms, particularly in binary classification tasks. Both F1 and **Intersection over Union (IOU)** values are within the range of 0 to 1, with higher values indicating superior performance. AUC, defined as the area under the Receiver Operating Characteristic curve, serves as a metric for evaluating the effectiveness of a binary classification model. Its value does not exceed 1, so a higher AUC value implies a closer alignment between predictions and ground truth labels. A value approaching 1 signifies a more accurate prediction in relation to the true labels.

Experimental Details. The CAFTB-Net was developed using the PyTorch framework [50]. Our experiments were conducted on a system running Ubuntu 16.04, utilizing two Nvidia GeForce GTX 2080 Ti GPUs in parallel to expedite model convergence during training. For training the network, we employed the adaptive moment estimation (Adam) optimizer [51]. The batch size for training images was set to 24, and the initial learning rate was configured at 1×10^{-3} . Scheduled reductions to 1×10^{-4} , 1×10^{-5} , and 1×10^{-6} were applied after the 30th, 60th, and 90th epochs, respectively. The forgery region of a document image is usually much smaller than the size of the image. This will produce more negative samples and cause the sample imbalance problem. To alleviate this

problem, this article uses two loss functions to supervise the training. They are Dice Loss [52] and BCE Loss.

Dice loss is derived from the Dice coefficient, a metric function utilized to measure the similarity between two samples. The Dice coefficient yields values within the range of 0 to 1, where higher values signify a greater degree of similarity between the two samples. The Dice coefficient is defined as follows:

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|}, \quad (9)$$

$|X \cap Y|$ denotes the common elements shared by sets X and Y , while $|X|$ and $|Y|$ correspond to the total number of elements in X and Y , respectively. To ensure that the denominator remains within the $[0, 1]$ range through successive iterations, the numerator is multiplied by 2. Considering that the forgery pixels have binary nature, the value is 1 if it is a positive sample; otherwise, it is 0. Therefore, this article incorporates the BCE loss. The final loss function is

$$L_{\text{Loc}} = \alpha L_{\text{Dice}} + \beta L_{\text{BCE}}, \quad (10)$$

where the weights α and β are set to 0.7 and 0.3, respectively.

Comparison Scheme. To show our algorithm's state of the art, this article compares it with some representative image forgery detection algorithms. For example, TSDIFL-Net [26] proposes a two-stream network to extract the RGB and noise domain features separately. CFL-Net [17] uses constrained convolution to learn image anomalies and locate forgery regions based on the anomalies. RRU-Net [53] is a U-Net based residual ring network for image stitching forgery detection, and its core idea is to enhance feature learning using residuals. MVSS-Net [54] utilizes semantic-independent image noise distribution and boundary features to locate forgery regions accurately. The SegFormer network [45] is a semantic segmentation network and is used to compose the NFEB of CAFTB-Net. RDIFL-Net [28] addresses the forgery image post-processing and channel transmission distortion perspective to make the forgery detection algorithm more robust.

4.2 Comparison against SOTA Methods

This section compares CAFTB-Net with the SOTA algorithm mentioned above. Since the forgery images detected by the original article of the compared algorithms are not document-like images, and the feature differences between natural and document images are too large, the pre-training weights provided by the original article of the compared algorithms are not used for fine-tuning but are re-trained. All algorithms use the same dataset. Furthermore, the experimental setup of the comparison methods is set up according to the description of the original article corresponding to the algorithms. It is worth noting that TSDILF-Net and RDIFL-Net are not provided with the code of the original articles, so we use the data provided in the original articles to compare the SACP dataset.

The comparative experimental results presented in Table 1 clearly demonstrate that CAFTB-Net has made significant progress in the field of document image forgery detection. In the SACP dataset, although the recall metric is slightly lower than MVSS-Net, all other metrics achieved the best results, with AUC, F1, and precision values of 0.933, 0.819, and 0.792, respectively. These values represent improvements of 14.6%, 1.5%, and 3.7% over MVSS-Net. The reason for the lower recall compared to MVSS-Net could be attributed to the higher number of incorrect pixel predictions in MVSS-Net, which may have resulted in greater coverage of tampered areas, leading to a decrease in precision. Overall, CAFTB-Net outperforms MVSS-Net in terms of overall performance, particularly in sample differentiation and prediction accuracy. SegFormer also achieved commendable results; however, its performance in all metrics falls short of CAFTB-Net. This discrepancy may stem

Table 1. Performance of CAFTB-Net with Other SOTA Algorithms on SACP and ICDAR Datasets

		TSDIFL- Net [26]	CFL- Net [17]	RRU- Net [53]	MVSS- Net [54]	SegFormer [45]	RDIFL- Net [28]	CAFTB- Net
SACP	AUC	-	0.760	0.653	0.814	0.773	0.837	0.933
	F1	0.30	0.752	0.636	0.807	0.749	0.552	0.819
	IOU	0.19	0.347	0.332	0.50	0.344	0.410	0.50
	Precision	0.20	0.77	0.613	0.764	0.754	-	0.792
	Recall	0.63	0.734	0.661	0.856	0.742	-	0.849
ICDAR	AUC	-	0.609	0.671	0.619	0.652	-	0.764
	F1	-	0.743	0.575	0.762	0.877	-	0.948
	IOU	-	0.094	0.266	0.126	0.200	-	0.417
	Precision	-	0.715	0.626	0.874	0.86	-	0.963
	Recall	-	0.773	0.531	0.717	0.893	-	0.934

Bold text indicates the maximum value of the row. “-” indicates that this result is not available.

from SegFormer’s limited ability to capture local image features, which are crucial for recognizing modification traces and non-natural image boundaries. The proposed method’s transformer branch structure draws inspiration from the SegFormer network, and its extraction of global features from the noise domain combined with spatial features enhances performance. Although CFL-Net and RRU-Net were initially designed for spliced image localization, they also achieved substantial F1 scores and precision when applied to document image datasets; however, the IOU scores from their testing were suboptimal at 0.19 and 0.332. IOU represents the intersection of predicted and tampered areas, indicating the limitations of these methods in accurately predicting tampered regions in document images.

The ICDAR dataset is characterized by a large amount of printed or handwritten text and smaller tampered areas, presenting a significant challenge for forgery detection tasks. The proposed method achieved the best results across all metrics in the ICDAR dataset, with scores of 0.764, 0.948, 0.417, 0.963, and 0.934, surpassing other comparative methods. This indicates that even in images with dense text and small tampered areas, the proposed algorithm effectively detects and localizes tampered regions.

To facilitate a more intuitive performance comparison between CAFTB-Net and the comparison algorithms, Figure 11 displays the detection and localization results for several forged document images. As can be seen from Figure 11, all the algorithms can roughly predict the forged regions of document-type images. In contrast, in predicting book covers, such as the third row of Figure 11, SegFormer and CFL-Net did not predict the forged regions completely. Meanwhile, SegFormer, RRU-Net, and CFL-Net have high false alarm rates on the samples in the SACP dataset, and there are also a large number of incorrectly predicted pixels on the ICDAR dataset. In particular, the RRU-Net algorithm failed to detect forged regions within the images from the ICDAR dataset, which shows that this algorithm is unsuitable for document image forgery detection and localization. In contrast, CAFTB-Net shows excellent and stable performance on both SACP and ICDAR datasets.

4.3 Robustness Analysis

In the Metaverse, when images are uploaded, they often undergo network transmission. This process may subject the images to effects such as JPEG compression and noise. Therefore, a forgery detection and localization framework that maintains robustness against post-processing operations is vital. In this article, we evaluate the network’s robustness against three specific post-processing operations,

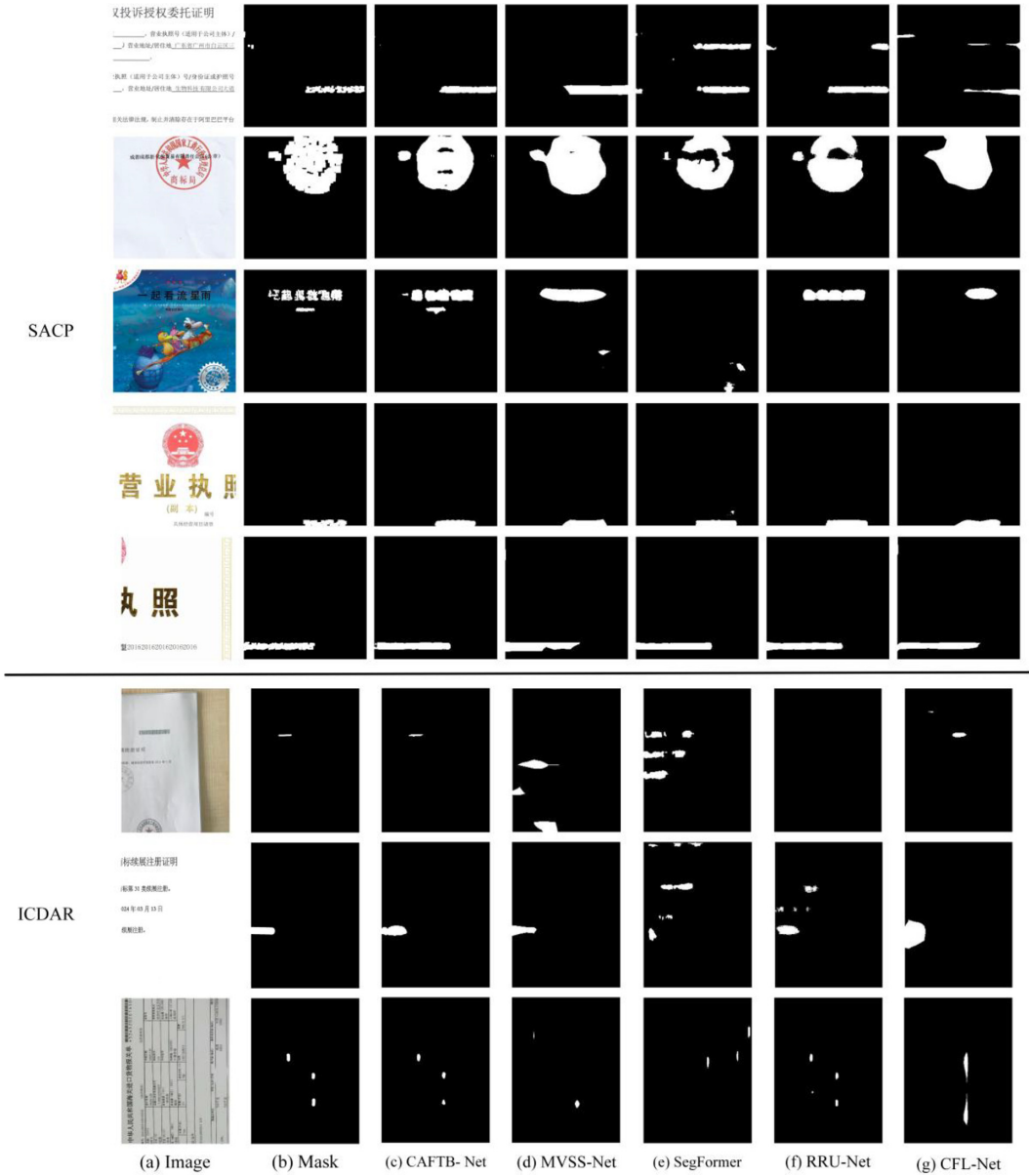


Fig. 11. Example of detection results of CAFTB-Net and the SOTA algorithm on the SACP and ICDAR dataset.

namely JPEG compression, median filtering, and Gaussian noise, and the specific experimental results are shown in Figure 12. Since the three algorithms RRU-Net, MVSS-Net, and SegFormer show better performance in the experiments in the previous section, these three algorithms are chosen in this subsection to compare with CAFTB-Net, and all the experimental results are obtained based on the SACP dataset.

The JPEG compression algorithm that removes redundant information and high-frequency details from an image, which may diminish the visibility of forgery traces in document images.

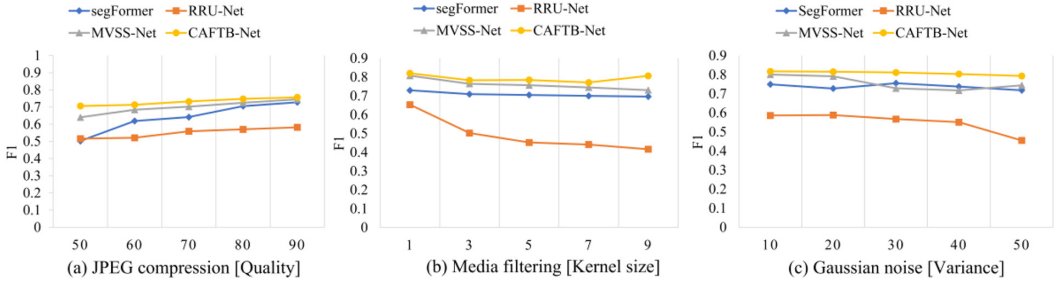


Fig. 12. F1 of CAFTB-Net and compared algorithms under JPEG compression, media filtering, and Gaussian noise attacks.

This experiment compressed document images in SACP using five JPEG quality factors ($QFs = \{50, 60, 70, 80, 90\}$). Then, the compressed images were used to evaluate the robustness of CAFTB-Net and the comparison algorithms. From Figure 12(a), it can be seen that all the detection algorithms degrade their detection performance when faced with low quality compressed images. The CAFTB-Net, MVSS-Net, and RRU-Net show more stable performance against JPEG compression. The SegFormer algorithm is more affected by JPEG compression.

Compared with the above three algorithms, CAFTB-Net demonstrates higher accuracy and stability. The reason is that JPEG compression may introduce global artifacts or noise, and these anomalous traces may appear in the whole image and are not only limited to the forgery region. CAFTB-Net improves the performance of document image forgery detection and localization through local feature extraction, global feature mining, and cross-domain feature fusion, and it can effectively deal with the feature changes and artifacts caused by JPEG compression.

Median filtering works by smoothing the image and removing noise. It blurs details and edges. So it is often used in document image forgery to blur forgery traces, making them more challenging to recognize. In the experiments of this subsection, median filtering is used to evaluate the robustness of CAFTB-Net and the comparison algorithm. In particular, the images in the SACP dataset are filtered using Python's Albumentation image processing library with filter kernel sizes $\{1, 3, 5, 7, 9\}$. The experimental results are shown in Figure 12(b), from which it can be seen that as the filter kernel increases, the image becomes blurrier and blurrier, and the detection performance of all the algorithms decreases. But it is smaller than the effect of JPEG compression. SegFormer and RRU-Net are more sensitive to the median filter processing, and the forgery localization performance decreases significantly when the filter kernel sizes are 3 and 5. MVSS-Net also exhibits a more significant fluctuation when the filter kernel is 5. Compared with the comparison algorithms, CAFTB-Net shows stable performance and high accuracy for median filtering. Since median filtering can blur the details and edges of the image, transformer's global feature modeling capability is more helpful for more accurate detection of forgery regions.

This article evaluated the model's resistance to noise. To achieve this, we added Gaussian noise with variances ranging from 10 to 50 to the test images. The evaluation results of various methods under the addition of noise are shown in Figure 12(c). It can be observed that, except for CAFTB-Net, the other methods are affected by the addition of noise, with RRU-Net experiencing the most significant performance decline. The data indicate that the proposed method demonstrates ideal robustness against Gaussian noise.

In order to more intuitively represent the advantages of the method proposed in this article against various post-processing operations, we visualize the results of various attacks, as shown in Figure 13. The visual results show that the proposed method delivers more precise prediction regions

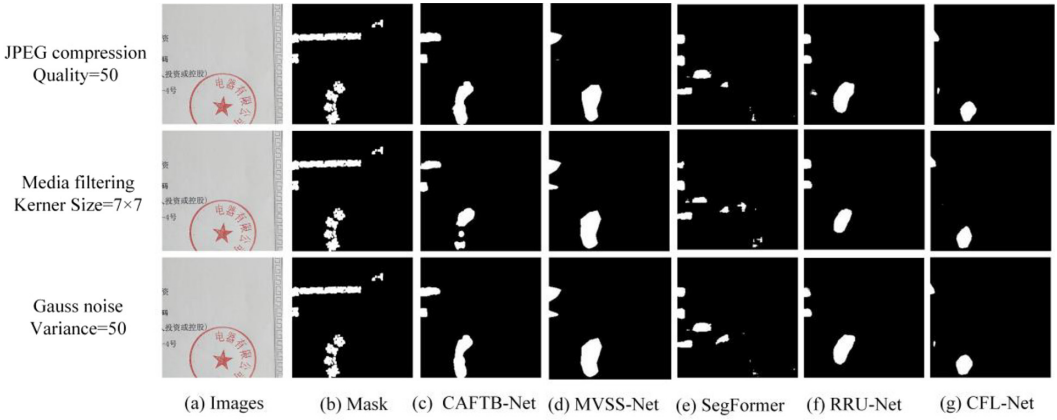


Fig. 13. F1 of CAFTB-Net and comparison algorithms under median filtering with different filter kernel sizes.

Table 2. Comparative Analysis between CAFTB-Net and Single-Branch Network, and the Impact of CAFM on Results, Which Were Obtained on the SACP Dataset

	AUC	F1	IOU
SIEB	0.7748	0.7311	0.3495
NFEB	0.7729	0.7494	0.3442
SIEB + NFEB	0.8096	0.7883	0.4490
SIEB + NFEB + CAFM	0.8326	0.8193	0.4988

Boldface text indicates single-column optimal results.

when dealing with post-processing operations. It maintains strong performance, underscoring its exceptional robustness.

4.4 Ablation Study

This subsection conducts ablation experiments for each branch network and CAFM in the CAFTB-Net to illustrate that the proposed dual branch network structure improves the performance of forgery localization compared to that using a single branch. In this experiment, the evaluation of SIEB's performance involves the use of SIEB and decoding modules. Likewise, NFEB's performance is assessed using NFEB and decoding modules. The CAFM is not included in the evaluation of either branch, with all other experimental parameter settings remaining consistent.

According to the experimental results in Table 2, the network with only SIEB or NFEB also achieves good forgery localization performance. However, the localization performance is better than both combining the two branches. The F1 score rises from 0.7311 and 0.7494 to 0.7883, which is improved by 7% and 5%, respectively. The IOU score significantly increases 28.5% and 30.4%, respectively, when using the two-branch network. This indicates that if only local features of CNN are used, the network focuses more on the document image content and does not extract global pixel correlation. And if only the global features in the noise domain are used, the local details of the document image will be lost. Therefore, combining CNN and transformer networks can improve document image forgery detection performance. In order to demonstrate the effectiveness

Table 3. Comparison of the Generalization Ability of CAFTB-Net

		AUC	F1	IOU
(SACP, ICDAR)	MVSS-Net	0.5215	0.2779	0.0170
	RRU-Net	0.5148	0.3700	0.0189
	CAFTB-Net	0.5500	0.6808	0.0296
(ICDAR, SACP)	MVSS-Net	0.5289	0.2217	0.0459
	RRU-Net	0.5174	0.1720	0.0385
	CAFTB-Net	0.5381	0.7112	0.0544

(SACP, ICDAR) indicates that the model is trained on the SACP dataset and then tested on the ICDAR dataset to obtain the results. (ICDAR, SACP) denotes that the model is trained on the ICDAR dataset and then tested on the SACP dataset to obtain the results.

Table 4. Running Times (Seconds) of Different Algorithms on SACP

Methods	Testing on 100 images	Testing on 1 image	Params
CAFTB-Net	9.7	0.09	91 M
CFL-Net [17]	14	0.13	112 M
RRU-Net [53]	3.2	0.03	4.1 M
MVSS-Net [54]	5.9	0.06	143 M
SegFormer [45]	7.1	0.07	8 M

of CAFM, we designed a comparison experiment between SIEB + NFEB and SIEB + NFEB + CAFM; from the experimental results of the two schemes, all evaluation metrics have been significantly improved, the AUC, F1 scores, and IOU scores have increased from 0.8096, 0.7883, and 0.449 to 0.8326, 0.8193, and 0.4988. There are 2%, 3%, and 11% improvement, respectively. This indicates that CAFM enables the network to focus on more effective forgery regions, fully integrates the local features of the air domain and the global features of the noise domain, and effectively reduces the false alarm rate for forgery region localization.

The generalizability of image forgery detection algorithms has been a challenge in forensics. This subsection evaluates the generalizability of CAFTB-Net. We choose two algorithms with better performance for comparison. The generalizability of the proposed algorithm and the comparison algorithm is evaluated on SACP and ICDAR datasets, the results are shown in Table 3.

4.5 Running Time

The operational cost of detection is also a key concern in practical applications. We tested the time taken to detect 100 images and 1 image on the SACP dataset, comparing it with other methods. The results are shown in Table 4. For a fair comparison, we standardized the batch size to 1 and conducted five tests for each method. After removing the longest and shortest runtimes, we averaged the remaining times to obtain the experimental results. The table below shows the runtime for five algorithms, where RRU-Net has the smallest parameter count and relatively fast runtime for both single image and batch processing. Although CAFTB-Net's detection speed is not optimal, its runtime is still reasonable, averaging 0.09 seconds per image.

5 Conclusions

This article addresses the security issues of document images within the Metaverse by proposing the CAFTB-Net algorithm, which is designed to detect and locate forged regions within document images in the Metaverse. Specifically, CAFTB-Net employs a dual-branch network to address the challenge of distinguishing between positive and negative samples by incorporating noise domain features to complement spatial visual features. Furthermore, a CAFM is designed to fuse spatial information and noise features, enhancing the model's generalization ability and robustness to predict more accurate results. Experimental results demonstrate that CAFTB-Net outperforms state-of-the-art methods on the SACP dataset and the 2023 Text Forgery Detection Competition (ICDAR). This indicates a significant performance improvement of our algorithm in handling the task of document image forgery localization.

Due to dataset limitations, the method proposed in this article cannot yet detect all types of document image tampering, and the issue of generalizability remains one of the challenges plaguing current forensic research. In the future Metaverse, forgery techniques will continue to evolve. Therefore, it is encouraged that more scholars join forensic research to develop detection methods with greater universality. Additionally, with the advancement of information transmission technologies, forgery techniques are no longer confined to images alone. Consequently, it is necessary to consider various carriers of information such as text, audio, and video to develop multimodal forensic methods. This will further propel progress in the security domain of the Metaverse and contribute to the establishment of a safer and more trustworthy digital world.

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Conflict of Interest

The authors declare that they have no conflict of interest.

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